Powder Coating

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Powder coatings are the ideal way to combine individual color creation and good resistance to environmental influences.

ALUPCO Co. “one of our main suppliers” has a State of The Art powder coating line; with pre-treatment for aluminum and steel, three powder booths and a fully automated curing oven. The whole production process is quality controlled and quality assured by a batch by batch- laboratory test on physical specimens.

WHY POWDER COATING?

Among the important factors influencing the choice of powder coating are the following:

* The Process simplicity, and the case automation of the coating lines.
* The fairly uniform and higher film thickness, highly decorative and protective.
* The film strength of the coat, and its durability.
* The attractive wide range of colors.

The Polyester TGIC Powder coating is ideal to protect metal surfaces for either indoor or outdoor architectural usage.

The powder is formulated from durable polymers, and curing agents and applied with electrostatic equipment.

SURFACE PREPARATION:

Surface preparation prior to powder coating is an essential step to ensure proper coating performance.

Methods for preparing the surface are several; mechanical preparation, chemical preparation & conversion coating.

METAL CHEMICAL PREPARATION:

A very important point in powder coating is the metal pre-treatment. The right chemical pre- treatment is the guarantee for a solid adhesion of the film to the metallic surface.

CHEMICAL PRE-TREATMENT FOR ALUMINIUM:

Step 1: Alkaline Cleaning:

To remove dust and grease from the surface.

Step 2: Rinsing.

Step 3: Etching:

To clean the aluminum surfaces from any residual oxides.

Step 4: Rinsing.

Step 5: Yellow chromating:

Chemical Reaction to:

* Protect the surface against corrosion.

* Better adhesion of the paint film.

Step 6: Rinsing.

Step 7: Rinsing with DI- water.

Step 8: Forced drying.

QUALITY CONTROL:

* Technical properties for powder coating on aluminum:

Film Thickness (DIN 50 984 / ISO 2808)
60 to 90 microns.

Adhesion: (DIN 53 151 / ISO 2409)

* Cross-cut test:

Cross hatch rating (GT 0): 100% Adhesion.

Flexibility:

* Bend test: (DIN 53 152 / ISO 1519)

* Conical Mandrel:

No cracks appear at narrow areas.

* Cylindrical Mandrel:

Passes 4 mm mandrel diameter.

* Erickson cupping test: (DIN 53 156 / BS 3900E4)

Indentation depth up to 6 mm without film cracking.

Impact Resistance:

* Falling Ball Test: (BS 6496C16)

No cracking or detachment at height of 25 cm/ 0.908Kg

(~23 cm/ kg).

Film Hardness: (DIN 53 153 / ISO 2815)

* Indentation Resistance Test:

According to Buchholz: Minimum 80/ 90.

Drilling, milling & sawing test: (ASTM D 2794)

No flaking of coating.

Resistance To Weathering:

* Natural Weathering (Florida Test):
No chalking, excellent gloss retention and color stability after
One-year exposure at an angle of 45o to South.

* Accelerated Test
Q.U.V. accelerated weathering test: (ASTM G53 - 84)
Excellent gloss and color stability.

Resistance To Water Of Condensation: (DIN 50 017 / BS 3900F2)

* Humidity Test:

No blistering or loss of adhesion after 1000 hours.

Salt Spray Resistance: (DIN 50 014 / BS 6496)

No blistering or loss of adhesion after 1000 hours.

Mortar Resistance: (ASTM C 207)

After 24 hours at specific conditions, mortar is easily removed from the coating, resulting neither in loss of adhesion nor in surface marring.